



A flow-based statistical model integrating spatial and nonspatial dimensions to measure healthcare access

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ABSTRACT

Assessing access to healthcare for an entire healthcare system involves accounting for demand, supply, and geographic variation. In order to capture the interaction between healthcare services and populations, various measures of healthcare access have been utilized, including the popular two-step floating catchment area (2SFCA) method. However, despite the many advantages of 2SFCA, the problems, such as inappropriate assumption of healthcare demand and failure to capture cascading effects across the system have not been satisfactorily addressed. In this paper, a statistical model for evaluating flows of individuals was added to the 2SFCA method (hereafter we refer to it as F2SFCA) in order to overcome limitations associated with its current restriction. The proposed F2SFCA model can incorporate both spatial and nonspatial dimensions and thus synthesizes them into one framework. Moreover, the proposed F2SFCA model can be easily adapted to measure access for different types of individuals, over different service provider types, or with capacity constraints in a healthcare system. We implemented the proposed model in a case study assessing access to healthcare for the elderly in Taipei City, Taiwan, and compared the weaknesses and strengths to the 2SFCA method and its variations.

1. Introduction

Good access to healthcare that enables people to benefit fully from a healthcare system is widely recognized as an important facilitator of overall population health. Access to healthcare is influenced by many factors, e.g., the supply of healthcare services, demand for healthcare, the population's health status, demographic characteristics, socioeconomic status, and geographical impedance between a population and healthcare services.

The concept of access to healthcare is complex and thus difficult to define and measure. The seminal paper by Penchansky and Thomas (1981) proposed a useful definition by accounting for each of the following dimensions: accessibility, availability, affordability, accommodation, and acceptability. Recently, some published literatures have introduced a conceptualization of access to healthcare by describing broad dimensions (Levesque et al., 2013; Saurman, 2016). Joseph and Phillips (1984) classified access according to two categories, potential and revealed access. Potential access refers to a person's ease of

accessing these services based on existing conditions but does not warrant the utilization of the service. Revealed access, based on potential access, focuses on the actual use of services. Both types of access can be further classified into spatial and nonspatial access based on how the healthcare access is influenced by spatial factors (e.g. spatial location and travel distance) and nonspatial factors (e.g. socioeconomic status, or cultural background) (Luo and Wang, 2003; Wang and Luo, 2005; Wan et al., 2012; Bissonnette et al., 2012; Wang, 2012).

Healthcare access is a multidimensional concept and needs to account for both spatial and nonspatial dimensions simultaneously (Khan, 1992). Spatial access varies across space because neither healthcare providers nor populations are evenly distributed (Luo and Wang, 2003). A widely accepted measure to assess spatial accessibility, the two-step floating catchment area (2SFCA) method, has been developed by Luo and Wang (2003). A key feature of the 2SFCA method is its use of catchments that are specific to population and service locations rather than relying on artificially pre-defined regions.

With regard to nonspatial access, it is influenced by a wider

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selection of demographic and socioeconomic variables that affect healthcare access, and thus it is difficult to identify the needs for every socioeconomic group, since there are potentially hundreds of combinations. The healthcare needs incorporate the wider socioeconomic and environmental determinants of health, such as culture, religion, deprivation, housing, diet, education and employment, so that we can say that the healthcare needs of a population are constantly changing according to demographic and socioeconomic factors. Whilst direct measures of mortality and morbidity are generally preferred measures of healthcare needs (Morrissey et al., 2008; McGrail and Humphreys, 2009a), they are not readily available for small geographic areas. Despite the difficulty of quantifying healthcare needs in a single measure, doing so is central to any assessment of healthcare access which is focused on minimizing inequities. Although researchers are aware of the importance of both spatial and nonspatial dimensions in assessing healthcare access, often the two are studied separately among the limited literature (Wang and Luo, 2005; McGrail and Humphreys, 2009a; Field, 2000; Wang and Tormala, 2014).

Healthcare access may be conceived as the interface between potential users and healthcare resources, and is influenced by characteristics of those who supply as well as those who utilize the services. Measurement of spatial access over a healthcare network involves accounting for demand, supply, and network structure. Popular approaches are based on floating catchment areas (Guagliardo, 2004; Luo, 2004; Comber et al., 2011; Ngui and Apparicio, 2011); however, the methods may estimate demand inaccurately over the system because they were initially developed simply to overcome the regional boundary issue. Spatial interactions, known as spatial flows, refer to economic or demographic flows between different locations, and imply a complementarity between two places engaged in a supply–demand relationship which is subject to certain costs (Hayes and Wilson, 1971; Bennett et al., 1985). Spatial interaction models (Wilson, 1967, 1971; Boltzmann, 2008) (SIMs) are thus especially suited to study the spatial interaction between providers and population locations in a healthcare network.

According to the principle of equity in healthcare access, people should have equal access to health services, which in practice is not achieved, due to inequalities in the distribution of medical resources such as healthcare facilities and physicians. This paper thus focuses on the methodology for measuring potential access with integration of spatial and nonspatial dimensions to provide a summary measure of access to healthcare from the individual's perspective, and aims at identifying areas that are underserved or at risk of being underserved by healthcare services. Towards that goal, a flow-based statistical model was proposed in this study to estimate spatial flows to improve the 2SFCA method by capturing individuals' decision-making and cascading effects across the system, and to thus estimate healthcare demand much more accurately. A key feature of the proposed model, besides the simultaneous modelling of multiple spatial factors by the SIM, is the accounting of the nonspatial factor effect, as well as the interactions between spatial and nonspatial effects. The variations of the 2SFCA method have not yet incorporated information from spatial flows that represent a realistic pattern of behavior in terms of how individuals choose to access healthcare services. The accuracy of the proposed model compared to the 2SFCA method and its variations is analytically demonstrated, and a case study of healthcare access for the elderly in Taipei City, Taiwan is used to compare these approaches.

2. A statistical model in measuring healthcare access

2.1. Review of the 2SFCA method

The original 2SFCA method is closely related to the simple provider-to-population ratios (PPRs) method (Wang, 2012). PPRs provide a crude measure of access as a supply-to-demand match ratio in an area. However, PPRs are restricted to differentiating access using

fixed geographical or administrative boundaries, and they ignore both cross-border movement between boundaries and distance decay within boundaries (Luo and Wang, 2003; Wang, 2012). The 2SFCA method builds upon the framework of PPRs, but instead uses floating catchments that originate from provider and population locations to allow the catchment of each provider and population to float based on the distances between each pair (Luo and Wang, 2003). The size of the catchment is determined by a choice of maximum travel time (or distance), where all services (or populations) within that catchment are considered accessible and equally proximate to that particular population (or service provider), whilst all locations outside of the catchment are not accessible. As the name suggests, the final 2SFCA score is calculated using two connected steps:

Step 1: For each service (j), find all populations that fall within a threshold distance (d_0) and calculate the provider-to-population ratio, R_j .

$$R_j = S_j / \sum_{k \in \{d_{kj} < d_0\}} P_k \quad (1)$$

where S_j is the capacity of healthcare provider j and P_k is the population of area k .

Step 2: For each population (i), find all services that fall within a threshold distance (d_0) and the accessibility of area i , A_i , is to sum the PPRs from step 1.

$$A_i = \sum_{j \in \{d_{ij} < d_0\}} R_j = \sum_{j \in \{d_{ij} < d_0\}} (S_j / \sum_{k \in \{d_{kj} < d_0\}} P_k) \quad (2)$$

Step 1 of the 2SFCA method determines what population k of size P_k are located within the catchment border, d_0 , of each service provider j of volume S_j , thus defining the provider-to-population ratio R_j within a service catchment. Step 2 then assigns these service ratios to the population by determining which services are located within the catchment border, d_0 , of each population i , and aggregating the Step 1 scores to calculate an area's access. The only decision required in applying the 2SFCA method is the catchment size, d_0 , which is then applied at both Steps 1 and 2.

Despite its relatively straightforward nature and popularity, the method's major limitation is its dichotomous approach that defines a service provider inside a catchment as accessible and one outside the catchment as inaccessible, and several studies have attempted to improve it (Ngui and Apparicio, 2011; McGrail and Humphreys, 2009b, 2014; McGrail, 2012).

The enhanced 2SFCA (E2SFCA) is a variation that applies an impedance function to create weights for opportunities within the catchment area to account for decay of the willingness to travel as distance increases (Luo and Qi, 2009). By generalizing the distance decay effect as a term $f(d)$ (Wang, 2012), all measures in the 2SFCA method can be synthesized as follows.

$$A_i = \sum_{j \in \{d_{ij} < d_0\}} R_j \cdot f(d_{ij}) = \sum_{j \in \{d_{ij} < d_0\}} (S_j \cdot f(d_{ij}) / \sum_{k \in \{d_{kj} < d_0\}} P_k \cdot f(d_{kj})), \quad (3)$$

where the distance decay function, $f(d)$, across a catchment can be treated as a continuous function, a piecewise-defined function, or a hybrid of the two across a single zone.

2.2. Modelling framework

Since equity in healthcare is widely accepted as an important goal of public policy, minimizing inequality in healthcare access can help to close the gap. For improving access or service over the healthcare system, most work uses measures of the accessibility of healthcare service to maximize service coverage, minimize travel needs of individuals, limit the number of facilities, maximize health, or combine some of these goals (Wang, 2012). There is a need to consider how cost-benefit analysis fits with considerations for equitable access to

healthcare.

One methodology of particular importance to healthcare access relates to how to model flows of individuals between population and healthcare locations, since these flows, known as spatial interactions, enable us to evaluate the demand for healthcare services (Boltzmann, 2008; Davies et al., 2013). Thus, to model flows that represent a realistic pattern of human behavior, namely how individuals choose to access services, both spatial and nonspatial perspectives must be taken into account. The probabilities of individuals' decisions to seek healthcare service generally form the first step in the process of estimating these flows. A well-justified assumption for our model was the framework of rational choice to model the most probable flows of individuals in a healthcare system. The concept of rational individuals implies both that their decisions are influenced by the relative merits of actions, for instance, the potential rewards available at different levels and locations of healthcare services, and that they seek to minimize costs, choosing nearby services and seeking to avoid waiting. From a modelling perspective, this implies that these factors should be included in any formulation of the utility associated with a particular location.

An individual's probability of utilizing services depends on the balance between the spatial and nonspatial access. In our model, the willingness to access healthcare is a function not only of distance, facility congestion including its size, and medical resources but also of incorporating an additional nonspatial factor, healthcare needs.

From the spatial perspective, access to healthcare services, particularly in regards to potential utilization behavior of the elderly considered in this study, may be affected by several factors, including appropriate supply, reasonable distance or time impedance to available services, and waiting time for getting treatment. Inadequate supply of healthcare services is the most obvious barrier to accessing services at times of need (Joseph and Phillips, 1984; McGrail and Humphreys, 2009a; Slifkin, 2002). The unequal distribution of medical resources contributes to the population's already unequal access to healthcare services. Since healthcare services are not ubiquitously available (Rice and Smith, 2001), distance and time separation between population and health service locations represents an important access barrier for elderly populations (Joseph and Bantock, 1982). Waiting time is measured to reflect the degree of congestion at a healthcare service, which represents the resources available. For the nonspatial perspective, healthcare needs represent an integral element of an individual's health state. Since achieving equitable access is widely accepted as an important goal of health service planning (McGrail and Humphreys, 2009a; Culyer and Wagstaff, 1993; Oliver and Mossialos, 2004), it follows that the burden of healthcare needs is directly related to a population's need. That is, a population with an increased burden of health (poorer health) will also have an increased need for access to healthcare. Several key factors used in this study are summarized in Fig. 1.

Initially, quantitative methods were employed to explore characteristics of factors measured by proxies affecting individuals' decision-

making for seeking healthcare services. The likelihood of an individual accessing a service provider depends upon the relative attractiveness of that provider. To model rational individuals' decisions, some concept of the attractiveness of a service provider is required. This can be formulated using a cost-benefit structure, as is normal for SIMs, where benefit represents the potential reward to a visit within the catchment of a healthcare service, and cost embodies a set of measures of the impedance of the interaction between each pair (Davies et al., 2013). To construct the proposed model, the probabilities of affecting the individual decision were developed from both spatial and nonspatial perspectives before integrating them to estimate the potential demand.

2.3. Modelling spatial probability using spatial interaction model

With the aim of exploring the spatial patterns of individuals' behavior of access to healthcare, we assigned individuals to sites of healthcare services using an entropy-maximizing SIM (Wilson, 1970). The purpose of this model is to estimate the most probable flows in a spatial system such as ours, given certain constraints.

Several elements are defined for our formulation. The SIM in this study was defined across a discrete system of two entities: population and healthcare locations. These are indexed by i and j respectively and embedded in space. To model the attractiveness of a healthcare service, we assume that the benefit for the healthcare service j is given by the logarithm of Z_j . The $\log Z_j$ is a measure of relative value, which is the ratio of the number of healthcare services in the catchment of healthcare service j to the average across the system to reflect diminishing returns to scale. For b_{ij} , the benefit of healthcare service j as perceived by individuals in area i , we therefore have

$$b_{ij} \approx \log Z_j, \forall i. \quad (4)$$

In the aspect of congestion for healthcare services, the accurate measurement is complex and difficult. In this study, the primary gauge by which an individual assessed whether the situation at a healthcare site j was congested was the probability of the waiting time, determined by the relative numbers of the total population in the catchment of j to the average across all provider sites. A low perceived chance of waiting time encourages access. Several studies for the waiting time have been proposed (Xavier, 2003; Brekke et al., 2008; Oche and Adamu, 2013); in this case, we took an adapted version of the queuing theory as our starting point:

$$P(\text{wait at healthcare } j \text{ to getting treatment in one time unit}) = 1 - \exp\left(-\frac{N_j}{M}\right) \quad (5)$$

where N_j is the population in the catchment of healthcare j , M is the average population across catchments of all healthcare sites. On the basis that increased probability corresponds to increased congestion, we therefore expressed congestion thus:

$$\text{congestion} \approx \frac{N_j}{M}. \quad (6)$$

We then also incorporated a linear function of the distance between i and j , as is typical for analogous healthcare systems (Wang, 2012; McGrail and Humphreys, 2009b). Taking this as proportional to d_{ij} , the distance between the locations of i and j , we could then combine it with (4) and (6) to obtain the full expression for cost-benefit structure:

$$w_1 \log Z_j - w_2 d_{ij} - w_3 \frac{N_j}{M}, \quad (7)$$

where the w_k , $k = 1, 2, 3$, were constants.

Following the standard entropy maximizing derivation of the SIM (Boltzmann, 2008; Davies et al., 2013; Wilson, 1970), the associated attractiveness measure of visiting healthcare j from area i , W_{ij} , can then be written as follows:

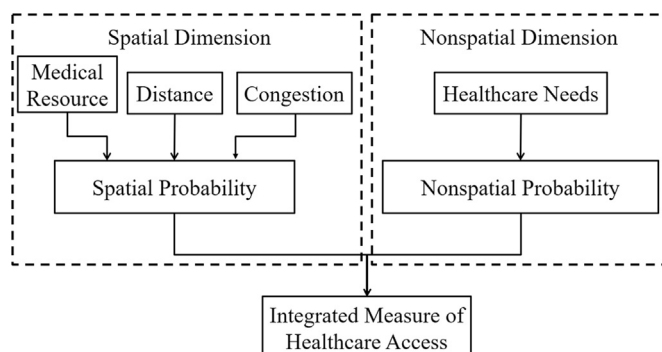


Fig. 1. Key factors affecting healthcare access.

$$W_{ij} = Z_j^\alpha \exp(-\beta d_{ij}) \exp\left(-\gamma \frac{N_j}{M}\right), \quad (8)$$

where α , β and γ are parameters to be obtained in calibration with real data. It was through the form of (8) that an exponential distance decay, discussed in the previous section, featured in the model. Motivated by the rational hypothesis, consistent with the decision-making progression of the individuals, we propose a probability model using the attractiveness function. From the perspective of the spatial dimension, the probability of an individual in area i choosing to access healthcare service j , p_{ij}^S , can be given by

$$\begin{aligned} p_{ij}^S &= P(\text{access from } i \text{ to } j) \\ &= P(\text{access from } i \text{ to the catchment of } j) \\ &\quad \times P(\text{access to } j \mid \text{access from } i \text{ to the catchment of } j) \\ &= \frac{\sum_{h \in \{d_{ih} < d_0\} \cap \{\text{services within the catchment of } j\}} W_{ih}}{1 + \sum_{h \in \{d_{ih} < d_0\} \cap \{\text{services within the catchment of } j\}} W_{ih}} \\ &\quad \times \frac{W_{ij}}{\sum_{h \in \{d_{ih} < d_0\} \cap \{\text{services within the catchment of } j\}} W_{ih}}. \end{aligned} \quad (9)$$

A logistic function is used for the first term in (9) to represent the existence of a threshold at which accessing healthcare in the catchment of healthcare j became appealing. The second term is a conditional probability to represent an individual in i choosing to access j , given that he chose to access healthcare in the catchment of j .

Intuitively, this probability will be small when the overall attractiveness of the catchment of healthcare j is low, whereas when the ambient level of attractiveness is high, the probability of accessing tends towards one. From another perspective, if an area is located within catchments of two healthcare services, then greater probability will arise in the more attractive of the two. For the spatial dimension, we therefore found the spatial access probability of an individual from an area to a designated healthcare service by using the SIM.

2.4. Measuring nonspatial probability by assessing healthcare needs

Population subgroups differ in terms of healthcare needs and access according to their age, sex, social class, ethnicity, and other nonspatial characteristics (Wang, 2012; McGrail and Humphreys, 2009a). Various population groups might also have different levels of healthcare needs. The significance of healthcare needs is also its association with service demand. The inclusion of health needs in a floating catchment area method is not new. In fact, it was a part of Wang and Luo's breakthrough paper (Wang and Luo, 2005) and first truly integrated by McGrail and Humphreys (2009a). Recently, a national-scale integration of spatial and nonspatial factors was also developed (McGrail and Humphreys, 2015). If a population has low healthcare needs, then the relative demand within that population is lower than indicated by only the raw population size. The aging of the population is a factor which is known to increase the needs and therefore the demand for healthcare services (Mechanic, 1999; Shrivastava et al., 2013). Most of the health problems of older age are associated with more than one chronic condition and require more frequent interactions with healthcare providers (Shrivastava et al., 2013). The elderly population is thus considered to be disadvantaged in the nonspatial dimension in this study. For the purpose of model-building, this implies a need to incorporate healthcare needs in the proposed mechanisms.

In order to reflect the true utilization behavior of the population, Ngui and Apparicio (2011) used the proportion of medical clinic users to weight the demand. However, as proposed by Yang et al. (2006), the effect of healthcare needs can simply be integrated within the 2SFCA method by its direct relationship to the service population size. Based on the elderly population in each area, we developed an healthcare needs indicator. However, the indicator needed to be transformed to an appropriate scale before its inclusion within the 2SFCA method to

estimate flows. The key transformation decision was to select a cap for areas of extreme healthcare needs (both high and low). If healthcare needs increase, then an individual's willingness to seek healthcare will increase, resulting in increased demand; whilst a minimum healthcare needs should be transformed to a positive value significantly less than one, which results in decreased demand. In this study, the healthcare needs of the elderly in area i was measured as the probability of affecting healthcare access in the nonspatial dimension, p_i^{NS} , and defined by

$$p_i^{NS} = P(\text{an individual in area } i \text{ needs healthcare service}) = E_i / \max_k \{E_k\}, \quad (10)$$

where E_i was the number of older people in area i , and $\max_k \{E_k\}$ represented the largest number of older people in all areas. This means that the higher the relative ratio of elderly population, the higher the probability of healthcare needs is; the lower the relative ratio of elderly population, the lower the probability of the healthcare needs is.

2.5. Integrating spatial and nonspatial probabilities within the 2SFCA method

The combination of the spatial and nonspatial probabilities yields the integrated probability to be considered for estimating flows of individuals to healthcare services. Thus we obtain:

$$p_{ij} = p_i^{NS} \times p_{ij}^S, \quad (11)$$

where p_{ij} is the probability that an individual in area i chose to access healthcare service j under both spatial and nonspatial considerations, p_i^{NS} is the probability that an individual in area i sought healthcare under the considerations of the nonspatial dimension, and p_{ij}^S is the probability that an individual in area i chose to access healthcare service j under the considerations of the spatial dimension.

By a careful examination, the nonspatial probability reflects a weighting factor for the demand side of healthcare, and thus can be considered along with the spatial probability. In other words, one may assign larger weights to population subgroups with high healthcare needs and directly incorporate this factor into the spatial probability.

We made an adjustment to allow individuals to have different probabilities in both spatial and nonspatial dimensions to different service providers. Although the E2SFCA aims to estimate the number of individuals that may potentially use a healthcare service, it is easy to extend the metrics to estimate the demand by multiplying the integrated probability. Under the assumption that decisions were independent between individuals, the number of individuals in area i that chose to access healthcare service j , N_{ij} , was therefore:

$$N_{ij} = P_i \times p_{ij}, \quad (12)$$

where P_i is the population size in area i .

The demand on a healthcare service as measured by Step 1 of the 2SFCA method was defined earlier in (1). If the demand increases, then the denominator will increase, resulting in a decreased provider-to-population ratio or R_j . Thus, for areas in the catchment of j of relatively high probability, N_{ij} should be increased, and for areas in the catchment of j of relatively low probability, N_{ij} should be decreased. Hence, Step 1 should be adjusted with the inclusion of the integrated probability for estimating the aggregate demand for healthcare j , and was given as follows.

Modified step 1 of 2SFCA:

$$R_j = \frac{S_j}{\sum_{k \in \{d_{kj} < d_0\}} N_{kj}} = \frac{S_j}{\sum_{k \in \{d_{kj} < d_0\}} P_k \times p_{kj}}. \quad (13)$$

The demand on a service as measured by Step 1 of the 2SFCA method was defined to calculate the population-to-provider ratio at each provider. In this study, the modified form of step 1 strived to improve the accuracy of 2SFCA by providing a more realistic repre-

sensation of access based on the spatial and nonspatial considerations of individuals. Rather than aggregating all populations that fall within a healthcare service catchment, the catchment population was estimated by the sum of all flows of people for each area within the catchment of a particular healthcare service. Therefore, we used provider-to-flow ratio instead.

The attractiveness of a particular healthcare location could be integrated within the 2SFCA method by its direct relationship to the population's catchment size (Step 2). The accessibility score of area i was then multiplied by the weighted attractiveness of healthcare j . Hence, Step 2 should be adjusted with the inclusion of the weighted attractiveness as shown in the following.

Modified step 2 of 2SFCA:

$$A_i = \sum_{j \in \{d_{ij} < d_0\}} \left(\frac{W_{ij}}{\sum_{k \in \{d_{ik} < d_0\}} W_{ik}} \right) \times R_j. \quad (14)$$

The original step 2 of the 2SFCA method regarded each provider-to-population ratio that fell within the population's catchment as contributing equally to the accessibility score. Instead of each healthcare service contributing equally, some healthcare services were more attractive than others. The step 2 of the 2SFCA method was thus modified as a sum of provider-to-flow ratios weighted according to the attractiveness of each healthcare service that fell within the catchment of a particular population location. We refer to the modified form of the 2SFCA method as the flow-based 2SFCA model (F2SFCA) in this study.

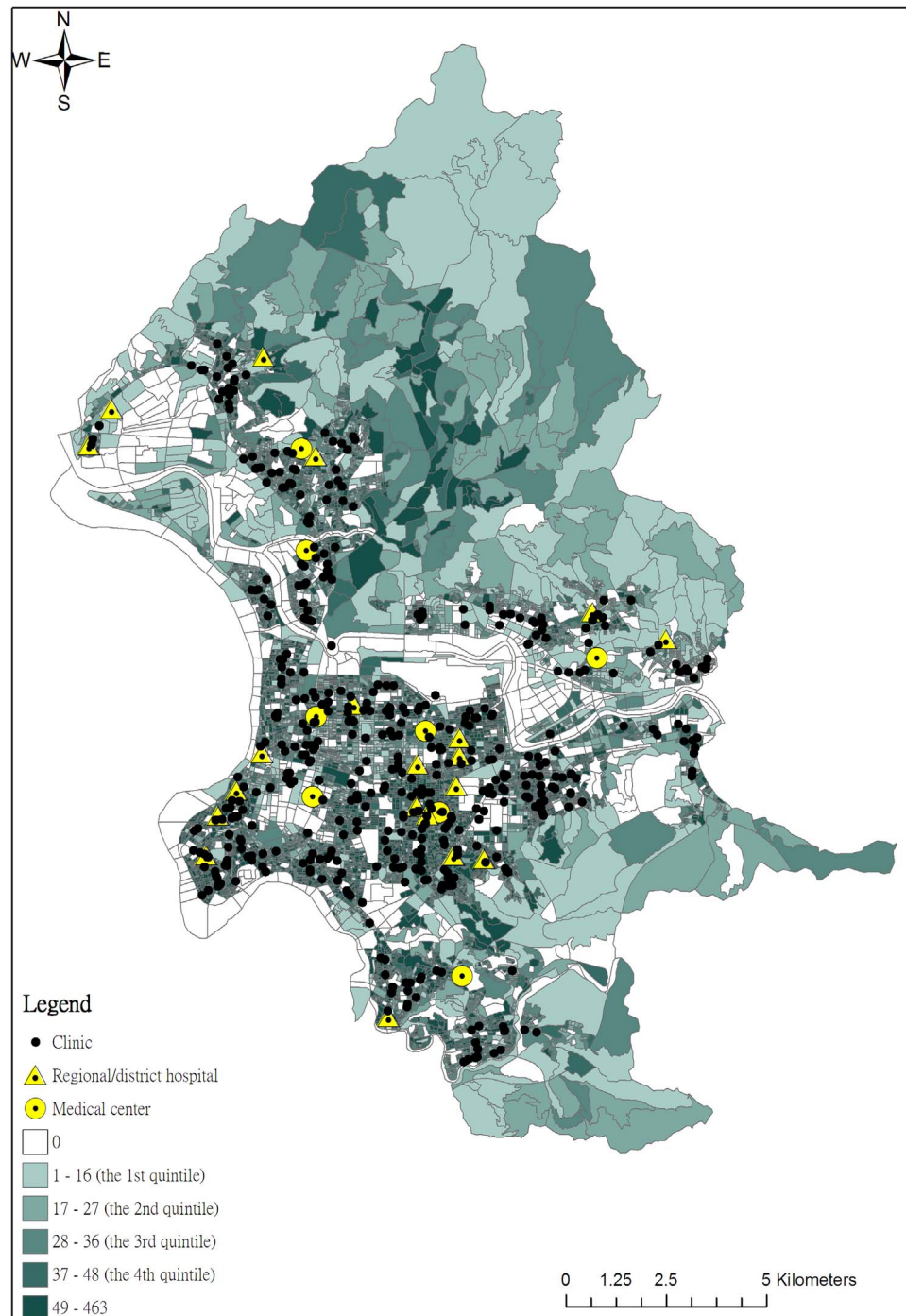


Fig. 2. The distribution of elderly population and healthcare services in Taipei City.

3. Demonstration case and data requirements

Today, the world is ageing. Older people already account for a sizeable proportion of healthcare patients, and as population ageing and chronic disease rates climb, that proportion will continue to increase. To ensure the later part of a person's life span is healthy, meaningful and dignified, easier access to healthcare services plays a critical role. Within Taipei City, in response to the aging of society, numerous programs have been introduced by the city government aimed at allowing the elderly to maintain an independent lifestyle and care for themselves, but access to healthcare may also be an influencing factor.

Fig. 2 depicts our study area. Taipei City is situated in the northern part of Taiwan. The city length from north to south is 27.655 km, and the width from east to west is 20.754 km. The perimeter of the city is 216 km, with a total area of 271.7997 square kilometers. Based on household registration, the population of Taipei City at the end of 2014 was 2,702,315, accounting for 11.53% of the total national population of 23,433,753 people, and the population density was about 9942 people per square kilometer. The elderly population (65 years and above) was 380,527, accounting for 14.08% of the population of Taipei City.

The first key data elements for measuring access to healthcare are both population and healthcare service locations and the calculated proximity between the two. Population size and location data in Taipei City were obtained from the Ministry of the Interior (MOI), using the smallest geographical unit, basic statistical areas (BSAs), where this study was based. BSAs are the smallest spatial units available in Taiwan used by the MOI as building blocks for collection, processing and output of population data. In 2014, a total of 11,847 BSAs were inside Taipei City's boundaries. The average elderly population size per BSA (excluding BSAs without residents) was approximately 34.21, and the average area of each BSA in Taipei City was 23,437.11 m². All boundary files were available from the MOI, and an approximate population location (center) for each BSA was calculated using the geographic centroid. The distribution of elderly population within BSAs of Taipei City is shown in Fig. 2.

Medical institutions which provide general practice (GP) service were used to represent healthcare services (providers) in this study. All records for service providers within Taipei City were collected from the National Health Insurance Administration (NHI), Ministry of Health and Welfare (MHW). The hospitals in Taiwan are accredited by MHW into three levels: medical centers, regional (district) hospitals and physician clinics. In this study, the capacity of each healthcare provider depended on its level and was proportional to the reciprocal of the number of different levels of healthcare providers. As of 2014, Taipei City had 568 medical institutions, including 8 medical centers, 20 regional/district hospitals and 540 clinics, and the distribution of these healthcare services also is shown in Fig. 2. Therefore the capacity of healthcare facilities based on hospital classification from medical center to clinic was in proportion 68:27:1. Locations of healthcare services were geocoded with the corresponding street addresses.

In our accessibility modelling, Distances between service locations and population (BSA) locations were calculated based on the road network of Taipei City using the Closest Facility tool of the Network Analysis module of ArcGIS software (ESRI Inc., Redlands, CA, USA). A maximum catchment size (both services and population) of 3 km was used. Four methods were compared, including the 2SFCA, the 2SFCA including the addition of the capacity of healthcare services, the 2SFCA method including the addition of the capacity of healthcare services and an exponential distance-decay function, and the proposed F2SFCA, by mapping out the accessibility scores of BSAs. In order to find out differences between the different approaches, quintile classification was used to break each access score data set into five categories to generate a choropleth map and then to compare indirectly with quintile rankings of each method. Access score in the top 20% were assigned a quintile

ranking of Category 5, the next 20% were assigned a ranking of Category 4, and so forth. In particular, comparisons were made in terms of relative positions of BSAs' accessibility scores in five pre-defined regions of interest (ROIs). The analysis in this study for geocoding and mapping calculated accessibility scores used ArcGIS and R Core Team (2014) software.

4. Results

As a first step towards construction of the model, we made specific observations of the healthcare system and the elderly in Taipei City. In seeking insight into individuals' decision-making during access to healthcare, an important task is to develop simplified and transparent proxy measures to underpin the proposed model. In this study, we considered the number of healthcare services relative to the mean across to all healthcare services as the proxy measure for medical resources in the catchment of a particular healthcare service; and the ratio of the elderly population to the number of healthcare services in the catchment of a particular healthcare service was the proxy measure for congestion. Given the high dimensionality of the model, many parameters of proxy measures were found to yield feasible results. As seen in Fig. 3(a), the population size of the elderly and the medical resources follow a clear nonlinear relationship. We fitted a nonlinear regression model with zero intercept to this data and the power was 0.65.

Analyzing the healthcare service locations further, we explore the pattern of the ratio of the elderly population to the number of healthcare services in the service catchment of a particular healthcare service, as seen in Fig. 3(b). The distribution corresponding to most

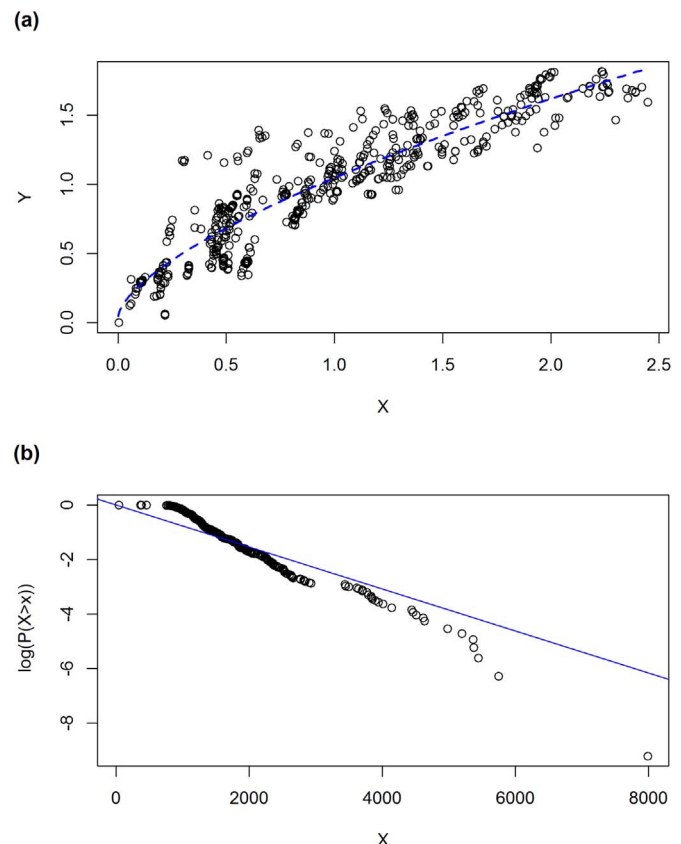


Fig. 3. Observations from data. (a) Relationship between the number of healthcare services relative to the mean across all healthcare services (X) and the number of the elderly in the service catchment relative to the mean across all healthcare services (Y). (b) Log-linear plot of the complementary cumulative distribution function of the ratio of the elderly population to the number of healthcare services in the service catchment (X). The straight line shows a hypothetical exponential distribution with parameter 0.00077.

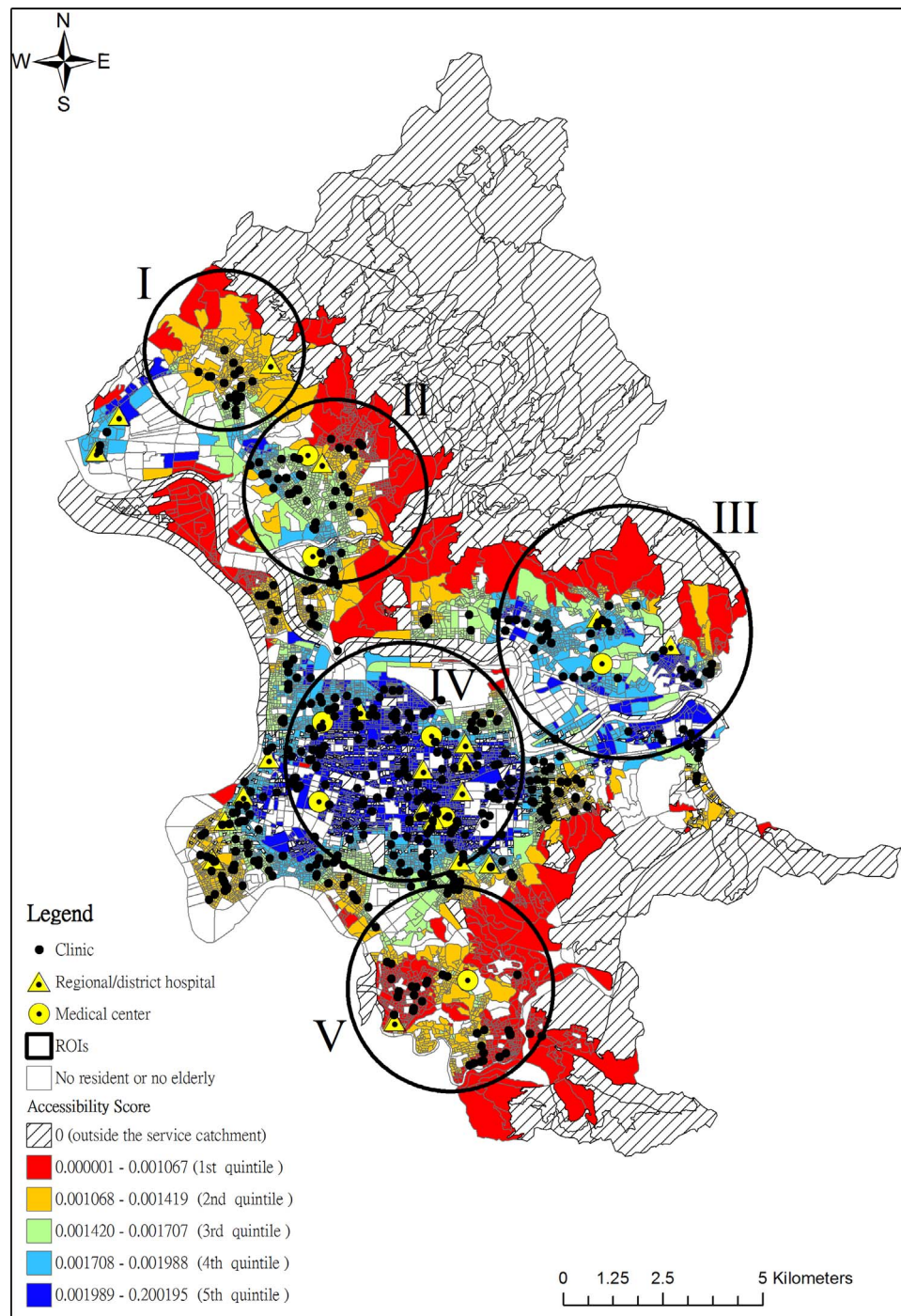


Fig. 4. Accessibility of healthcare service in Taipei City using the 2SFCA method.

common forms, and the best fit was provided by an exponential distribution with parameter 0.00077 (p -value < 0.001 and the adjusted R-squared = 0.9053). We also consider the distances the elderly travelled to the healthcare service locations to follow a distance-decay relationship.

Thus a parameter set chosen for the attractiveness function of (8) according to the features observed in the data was as follows:

$$\alpha = 0.65, \beta = 1 \text{ and } \gamma = 0.00077.$$

Next, access was estimated for the healthcare service network associated with the elderly in Taipei City. The proposed F2SFCA model was compared to some existing methods by mapping out the accessibility scores of BSAs. In particular, comparisons were made in terms of

relative positions of BSAs' accessibility scores in five pre-defined five regions of interest (ROIs) to the 2SFCA method and the Enhanced 2SFCA (E2SFCA) method.

Fig. 4 shows the results of applying the original 2SFCA method to measure access to healthcare in Taipei City. As expected, comparing this map against Fig. 2 confirms that accessibility is low in sparsely elderly populated areas and generally higher in more elderly populated areas where healthcare services are more likely to be located such as ROI IV. Not surprisingly, it is seen that by using the 2SFCA, there is a clear pattern of higher accessibility in BSAs within the catchment of services and low or non-existent accessibility outside the catchment.

Fig. 5 shows the change to accessibility scores seen in Fig. 4 after considering the capacity of healthcare services. BSAs exhibiting the

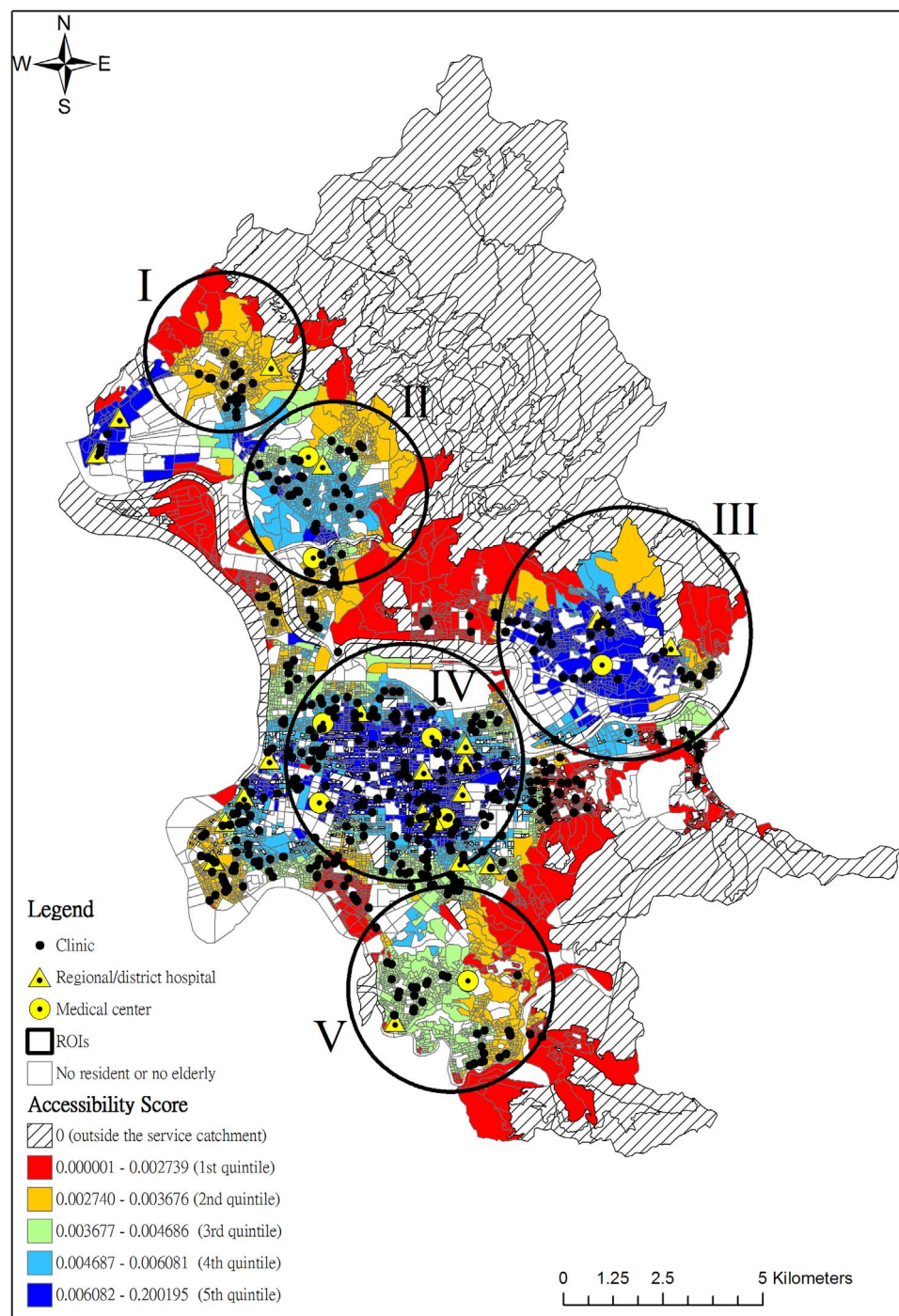


Fig. 5. Accessibility of healthcare service in Taipei City using the 2SFCA method including the addition of the capacity of healthcare services.

highest accessibility in Fig. 5 are characterized by an overlap from at least one of the large healthcare services within the 3 km catchment. These apparently high accessibility scores are because the 2SFCA method does not differentiate between impedance of 1 and 3 km, as illustrated in ROI II, III and IV of Fig. 5.

Fig. 6, which now includes the addition of an exponential distance-decay function (graduated between 0 and 3 km impedance), appears to more appropriately represent accessibility when compared to Fig. 5 (without the distance-decay function), since the function would ensure that small travel impedance between populations and available services has a high accessibility score. Generally, accessibility is highest in ROI II, III and IV of Fig. 6 where services are concentrated, and lowest in ROI I of Fig. 6, where there are fewer services. Thus, it appears that the

2SFCA method can be significantly improved by the addition of a distance-decay function to significantly account for impedance within a service catchment.

The resulting map from applying the proposed F2SFCA model of access to healthcare service in BSAs of Taipei City is shown in Fig. 7. This map has been calculated using a modified version of the 2SFCA method which included the addition of three spatial factors (medical resource, distance and congestion) to improve the measure of spatial accessibility as well as the integration of measures of nonspatial factor (healthcare needs). Several observations from Fig. 7 are noteworthy. As expected, the accessibility scores of BSAs around services are categorized into the highest score group, and especially with multiple services, as shown in ROI II, III and IV. Although there are high congestion

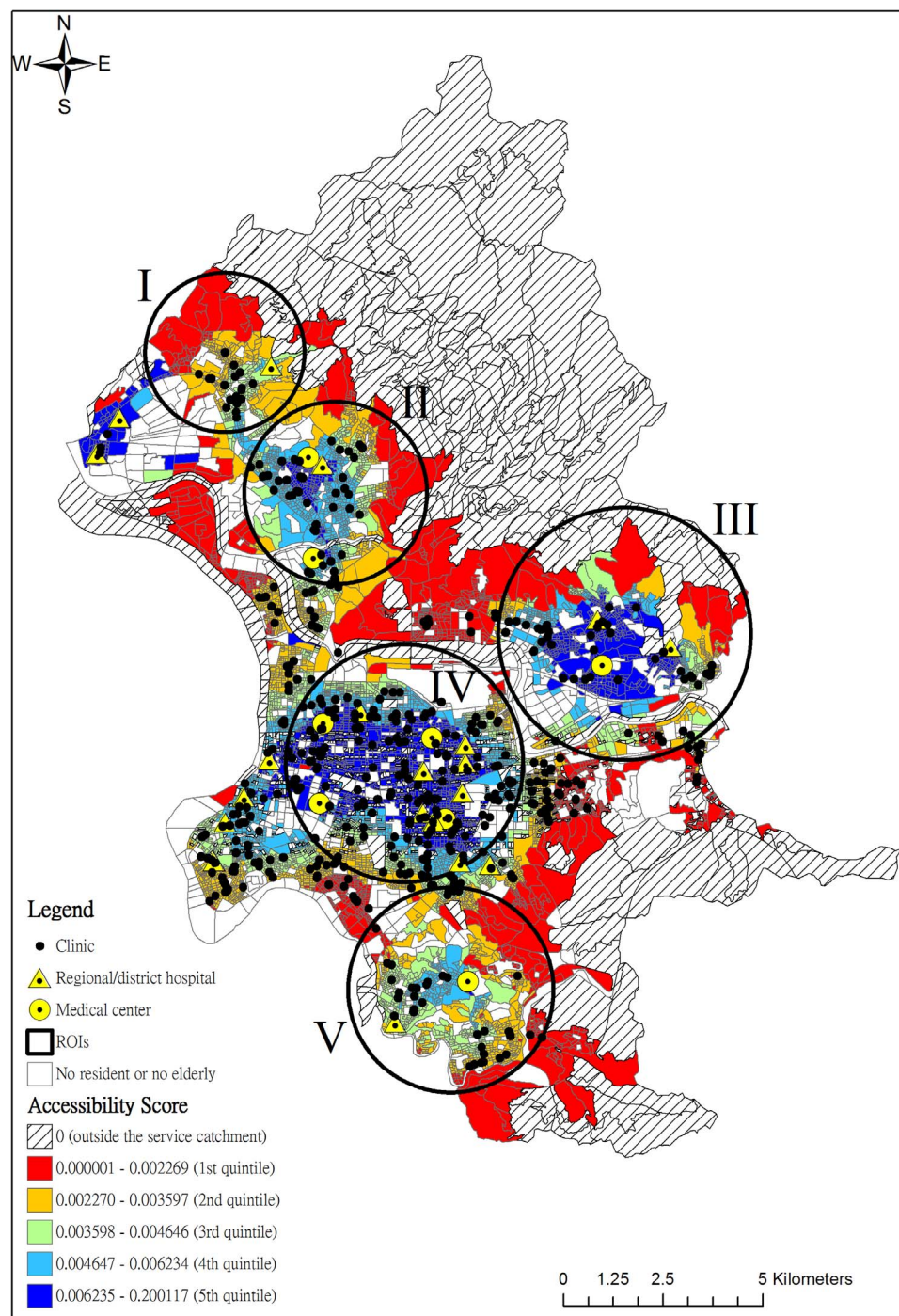


Fig. 6. Accessibility of healthcare service in Taipei City using the 2SFCA method including the addition of the capacity of healthcare services and an exponential distance-decay function.

(densely populated) and high healthcare needs in these high accessibility BSAs, they are usually rich in service resources and have short travel distances. Because the realized visits per service location estimated by the proposed F2SFCA method is higher than the E2SFCA approach on the demand side, and the number of facilities is not affected on the support side, the proposed F2SFCA method tends to deflate the number of BSAs that the 2SFCA method identified as having high accessibility in populated areas, as illustrated in ROI II and IV. By contrast, the 2SFCA and E2SFCA methods do not lose any elderly people. This is because the proposed F2SFCA method can capture the cascading effects by considering two additional spatial factors. For the same reason, this explains why the proposed F2SFCA method tends to show higher accessibility of BSAs around the service

boundary that the 2SFCA method identified as having low accessibility, as shown in ROI I, III and V.

In order to better understand the differences between different methods, the proportion of the elderly population and BSAs changes in five pre-defined ROIs have been tabulated. As shown in Table 1, most of the elderly population and BSAs in ROI I are placed in lower access categories (Category 1 and Category 2) using the 2SFCA and the 2SFCA method including the addition of the capacity of healthcare services. Compare with the E2SFCA method, the proposed F2SFCA method decreases the proportion of the elderly population and BSAs in Category 1. Table 2 reveals that the proportions of the elderly population and BSAs classified into the highest ranking category (Category 5) as assessed by the proposed F2SFCA method are less

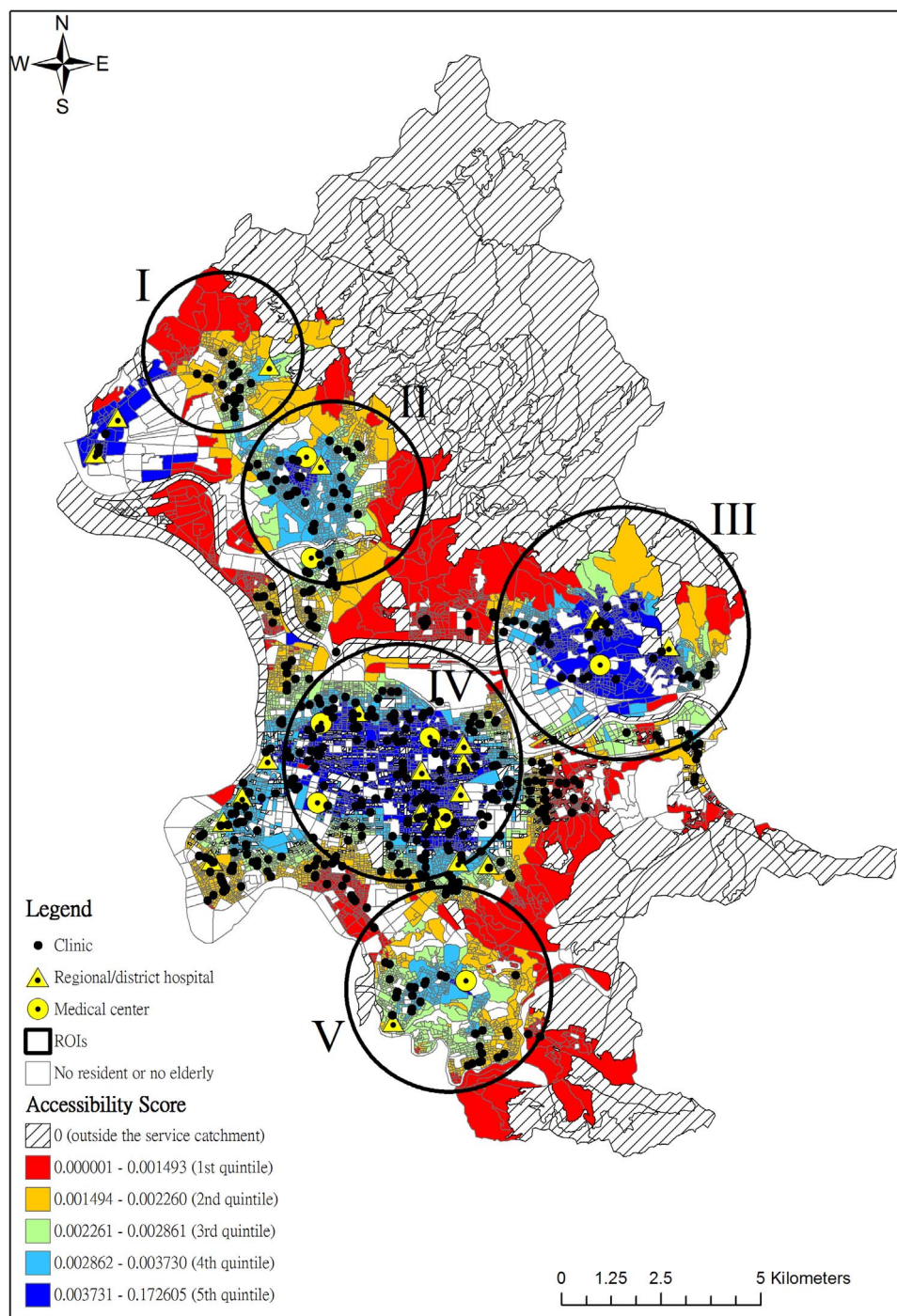


Fig. 7. Accessibility of healthcare service in Taipei City using the proposed F2SFCA model.

Table 1

The proportion of the elderly population and BSAs change by quintiles classification for the different methods in ROI I (Elderly population: 13129, BSAs: 393).

Quintiles classification (from low to high)	The 2SFCA		The 2SFCA method including the capacity of healthcare services		The 2SFCA method including the capacity of healthcare services and an exponential distance-decay function (the E2SFCA)		The F2SFCA	
	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs
Category 1	0.46%	1.27%	0.64%	1.78%	10.88%	11.20%	7.70%	8.65%
Category 2	69.56%	69.47%	80.24%	78.63%	42.06%	39.19%	43.57%	40.71%
Category 3	27.37%	25.70%	0.00%	0.00%	38.68%	40.20%	38.97%	39.44%
Category 4	1.44%	1.78%	17.63%	17.30%	6.89%	7.12%	8.27%	8.91%
Category 5	1.17%	1.78%	1.49%	2.29%	1.49%	2.29%	1.49%	2.29%

Table 2

The proportion of the elderly population and BSAs change by quintiles classification for the different methods in ROI II (Elderly population: 35511, BSAs: 1043).

Quintiles classification (from low to high)	The 2SFCA		The 2SFCA method including the capacity of healthcare services		The 2SFCA method including the capacity of healthcare services and an exponential distance-decay function (the E2SFCA)		The F2SFCA	
	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs
Category 1	19.33%	20.33%	0.80%	0.58%	2.28%	1.92%	2.12%	1.82%
Category 2	17.95%	18.22%	20.19%	20.42%	12.80%	13.61%	13.74%	14.29%
Category 3	46.56%	44.01%	23.44%	23.49%	20.02%	20.71%	25.47%	26.08%
Category 4	15.72%	16.87%	50.70%	50.53%	50.58%	49.66%	52.26%	50.91%
Category 5	0.44%	0.58%	4.86%	4.99%	14.32%	14.09%	6.41%	6.90%

Table 3

The proportion of the elderly population and BSAs change by quintiles classification for the different methods in ROI III (Elderly population: 27842, BSAs: 1099).

Quintiles classification (from low to high)	The 2SFCA		The 2SFCA method including the capacity of healthcare services		The 2SFCA method including the capacity of healthcare services and an exponential distance-decay function (the E2SFCA)		The F2SFCA	
	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs
Category 1	2.58%	3.55%	6.53%	6.73%	4.73%	4.55%	2.98%	2.91%
Category 2	6.81%	8.01%	12.03%	14.56%	11.65%	12.65%	7.82%	7.73%
Category 3	22.51%	20.84%	5.76%	5.91%	15.84%	16.47%	16.62%	17.65%
Category 4	41.02%	40.04%	13.55%	13.10%	20.58%	21.93%	19.32%	21.47%
Category 5	27.08%	27.57%	62.13%	59.69%	47.19%	44.40%	53.26%	50.23%

Table 4

The proportion of the elderly population and BSAs change by quintiles classification for the different methods in ROI IV (Elderly population: 100771, BSAs: 2688).

Quintiles classification (from low to high)	The 2SFCA		The 2SFCA method including the capacity of healthcare services		The 2SFCA method including the capacity of healthcare services and an exponential distance-decay function (the E2SFCA)		The F2SFCA	
	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs
Category 1	0.06%	0.07%	0.27%	0.30%	0.24%	0.22%	0.20%	0.19%
Category 2	1.10%	0.97%	4.69%	4.76%	7.04%	6.51%	6.88%	6.36%
Category 3	11.32%	8.97%	16.75%	14.43%	14.75%	13.24%	15.10%	13.43%
Category 4	24.79%	23.96%	30.63%	30.99%	26.62%	25.04%	24.77%	24.22%
Category 5	62.72%	66.03%	47.65%	49.52%	51.35%	54.99%	53.05%	55.80%

Table 5

The proportion of the elderly population and BSAs change by quintiles classification for the different methods in ROI V (Elderly population: 38596, BSAs: 1029).

Quintiles classification (from low to high)	The 2SFCA		The 2SFCA method including the capacity of healthcare services		The 2SFCA method including the capacity of healthcare services and an exponential distance-decay function (the E2SFCA)		The F2SFCA (proposed)	
	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs	% elderly	%BSAs
Category 1	56.51%	58.21%	8.47%	8.55%	10.74%	11.08%	8.43%	8.16%
Category 2	34.66%	33.43%	25.17%	27.41%	35.63%	35.37%	35.14%	35.57%
Category 3	8.83%	8.36%	64.96%	62.97%	34.78%	35.08%	33.93%	33.53%
Category 4	0.00%	0.00%	1.40%	1.07%	17.66%	17.10%	20.90%	20.99%
Category 5	0.00%	0.00%	0.00%	0.00%	1.19%	1.36%	1.60%	1.75%

than those as assessed by the E2SFCA method. The distributions of the proportion of the elderly population and BSAs changes in ROI III are showed in Table 3. In this table, the four methods are different in the distributions of the proportion of the elderly population and BSAs. Table 4 shows that there is no significant difference in the distributions of the proportion of the elderly population and BSAs. As shown in Table 5, the 2SFCA method has the opposite proportion distribution pattern to the others in ROI V.

5. Discussion

In this study, we aim to produce an integrated model which can give

rise to a more realistic representation of healthcare access by evaluating spatial flows. Motivated by the assumption of rational choice, consistent with the decision-making progression of the individual, we have proposed a statistical model, F2SFCA, to estimate flows of people according to the attributes of individuals and the choices the individuals made. Access to healthcare varies in space, since it is related to spatial and nonspatial factors. Our model draws on other factors in addition to distance; our contribution is in their combination and adaptation to produce an integrated model of healthcare access. The proposed F2SFCA model incorporates the fundamental features of seeking healthcare, and might be applicable to others of a similar nature.

The advantage of the proposed model is its ability to identify areas with access disadvantage at a much finer geographical scale in a more realistic way. This advantage has been achieved through incorporation of spatial and nonspatial factors. In the spatial dimension, an improved method to measure potential demand was developed by introducing the SIM to the 2SFCA method. The SIM was used to estimate the spatial access probability according to the attractiveness of a particular service provider. Additionally, an important nonspatial factor, healthcare needs, has been integrated within the proposed method in order to highlight the relationship between the nonspatial dimension and healthcare access. Then the product of spatial and nonspatial access probabilities was used to estimate the potential demand for a particular service provider accurately and embedded within the 2SFCA method to calculate the provider-to-flow ratio. All data have been collected and measured using the smallest available geographical unit (BSA), which enables the proposed model to measure access at a much finer scale than alternative measures. As a result, application of the proposed model is likely to achieve a more equitable distribution of healthcare services.

For many catchment models (Luo and Wang, 2003; McGrail and Humphreys, 2014; Luo and Qi, 2009; Luo and Whippo, 2012), the estimated accessibility measure increases when more service choices are available to a population. However, the proposed model estimates the cost-benefit of potential access, and this does not increase accessibility if a new choice is congested or inconsequential. Hence, it is an appropriate method for measuring accessibility for any healthcare service where populations are presented with a choice of services with overlapping service catchments.

The ability to quantify potential access over the healthcare system by using the 2SFCA method has been recognized for some time; however, the majority of variations of the 2SFCA method have only focused on the spatial motives of individuals, ascribing willingness to travel distance for healthcare (McGrail and Humphreys, 2009b, 2014; Luo and Qi, 2009; Luo and Whippo, 2012). The proposed F2SFCA model increases the level of complexity by including other factors in addition to distance, and using real data to calibrate the parameters in the model to correctly represent a more realistic representation of access to services.

Within the 2SFCA method and its variation methods (Luo and Wang, 2003; McGrail and Humphreys, 2009b; Luo and Qi, 2009; Luo and Whippo, 2012; Langford et al., 2012), each service provider located within the population's catchment is treated as accessible and available. The access opportunities are equally likely or use a distance impedance function to create weights within the service catchment. However, it is complex to model how individuals chose service providers. For different individuals and provider choices, one can model the trade-off between the access cost and the access benefit of particular providers using cost-benefit analysis (Davies et al., 2013). In this study, the spatial access probabilities were represented by the relative attractiveness of particular providers (measured by proxies such as medical resources, distance and congestion) formulated using the cost-benefit structure. Thus the proposed F2SFCA model can provide a better way to capture changes in both demand side and supply side.

A crucial factor in assessing accessibility is an understanding of how, where and when individuals access services. Analytically, the total number of individuals captured by all healthcare services in the 2SFCA method is larger than the number of individuals expected based on population size (Li et al., 2015). The 2SFCA method realizes additional choices regardless of their relative competitiveness with existing choices. In contrast, the proposed F2SFCA model adds the cost-benefit structure to address the competition by service providers, since the willingness to visit is not only a function of distance but also one of service congestion and medical resources. As a result, the proposed model can capture cascading effects in the healthcare network, where a change in attractiveness of a particular healthcare service for one

population leads to different decisions and thus influences individuals in another location.

In this paper, only one nonspatial factor, health needs, was incorporated in the proposed F2SFCA method. A high health needs score (transformed to a probability value) effectively increases the size of the population 'competing' to access available services; a low health needs score effectively reduces a provider's catchment size. To achieve the goal of health for all and to eliminate financial barriers to healthcare services, the government of Taiwan launched the National Health Insurance (NHI) program to provide universal medical care coverage on March 1, 1995. Those who cannot afford their premiums are eligible for assistance from the National Health Insurance Administration (NHIA) to ensure that health expenditures do not discourage patients from seeking necessary medical help. Thus, financial and socioeconomic barriers were ignored in this paper. Citizens insured under the NHI program have the freedom to choose any contracted facilities/institutions for their healthcare need. Therefore, in Taiwan, healthcare utilization has become well determined by health needs (Wang and Yaung, 2013).

In this paper, we particularly focus on the elderly population, inasmuch as older people tend to utilize more health services due to their deteriorating health status and frequently experienced chronic health conditions. The burden of increased travel may affect the elderly significantly, especially for regular hospital visits. Using a piecewise function with multiple distance decay weights across a catchment, such as the generalized 2SFCA (Wang, 2012) or a Huff-Based Model (Jia et al., 2017), to reflect the burden of travel for the elderly may be more suitable. We leave this issue for further discussion in the future.

The paper adds to a growing number of studies focused on the issue of healthcare access. There are limitations with the proposed approach. The key limitation to this study is the lack of available empirical data on 'real' healthcare service access behavior and its relationship to geography. The proposed model assumes that individuals are trading off between cost and benefit rationally across a healthcare network, while in reality there might be many other factors considered by individuals. Several theories have been developed to explain adherence as a rational process, that is, the person making health decisions based on the balance of perceived costs and benefits. Notions of the rational patient evolved out of several psychological models including the health belief model (Harrison et al., 1992), protection motivation theory (Armitage and Conner, 2000), and theory of planned behavior (Albarracín et al., 2001). Research on healthcare access should focus not only on where the patient comes from, but also on what factors affect the process of decision-making and health-related behavior. Decisions are made deliberately by weighing perceived costs and benefits of choices. This is applicable in the healthcare access case, and its implications are therefore a well-justified basis for rational individuals. In addition, a dichotomous approach was used to calculate potential service demand for a particular provider by aggregating all population locations that fall within its catchment border, and this is assumed to be zero for anywhere beyond this. Limitations are associated with the use of the smallest available geographical unit as the unit of spatial analysis. To measure small area variations of access necessitates that all data be collected and measured using the smallest available geographical unit. Due to confidentiality reasons, exact addresses of individuals are suppressed and replaced with population locations. Results are also dependent on the specific decay function, proxies and parameters chosen in an attractiveness function.

6. Conclusions

This paper outlines the development of a statistical model of access to healthcare services designed to provide a better and more sensitive measure to healthcare access, particularly in its ability to identify areas with access disadvantage at a much finer geographical scale. Even when real utilization data are inadequate or unavailable, the proposed model

may have its uses in contributing to a logical framework for analyzing tendencies, and allows individual choice to balance resources, travel and congestion within a population catchment.

Fundamentally, the proposed F2SFCA model has a different framework than many catchment methods, since it focuses on estimating the potential access associated with flows of individuals that are affected by spatial and nonspatial factors. It may be most important to use the proposed model when a healthcare system has services with overlapping catchment zones, when one wants to capture the nuances of access across populations, or when one needs to develop interventions to improve access.

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